

LeWorldModel: Stable End-to-End Joint-Embedding Predictive Architecture from Pixels

<https://arxiv.org/abs/2603.19312>

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杨立昆和世界模型

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Yann LeCun

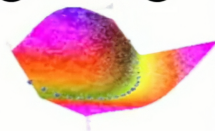
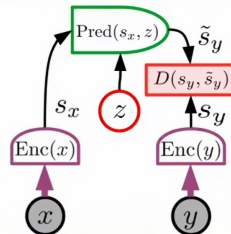
- ▶ 法国计算机科学家，图灵奖得主，卷积神经网络的重要奠基者。
- ▶ 1987 年前后，在多伦多大学从事博士后研究，导师 Geoffrey Hinton。
- ▶ 1988 年加入贝尔实验室 (Bell Labs)，开发卷积神经网络。
- ▶ 2013 年加入 Facebook，后任 Meta 首席 AI 科学家。
- ▶ 2018 年与 Geoffrey Hinton、Yoshua Bengio 共同获得图灵奖。
- ▶ 2025 年离开 Meta，创办 **AMI lab**，聚焦世界模型与类人人工智能。





Recommendations:

- ▶ **Abandon generative models**
 - ▶ in favor joint-embedding architectures
- ▶ **Abandon probabilistic model**
 - ▶ in favor of energy-based models
- ▶ **Abandon contrastive methods**
 - ▶ in favor of regularized methods
- ▶ **Abandon Reinforcement Learning**
 - ▶ In favor of model-predictive control
 - ▶ Use RL only when planning doesn't yield the predicted outcome, to adjust the world model or the critic.
- ▶ **IF YOU ARE INTERESTED IN HUMAN-LEVEL AI, DON'T WORK ON LLMs**



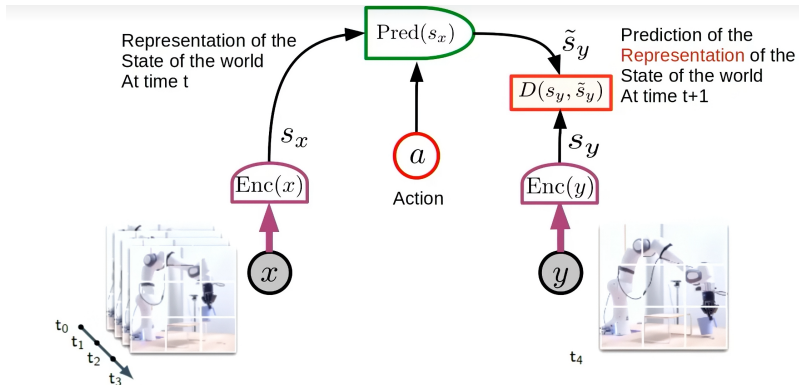
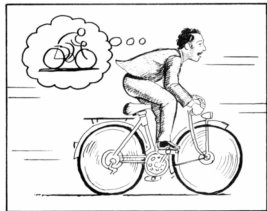


Future Universal Virtual Assistants

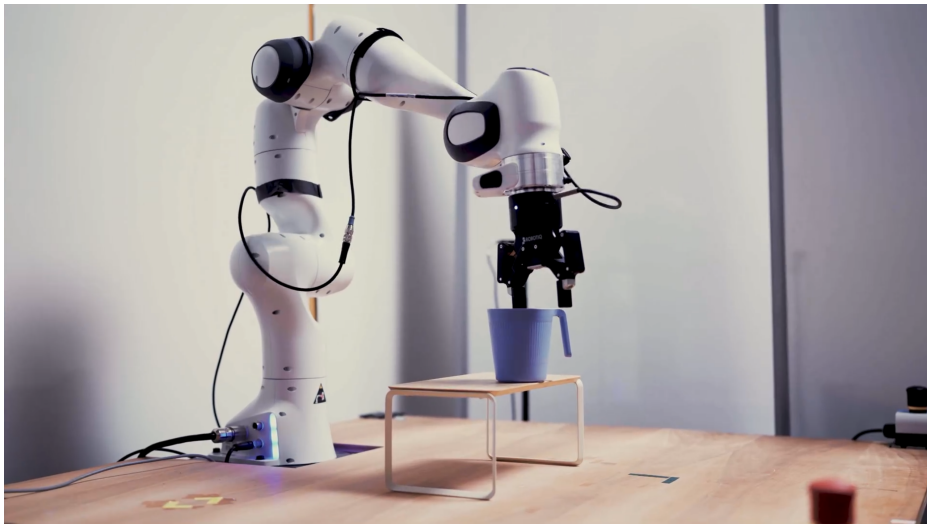
- ▶ All of our interactions with the digital world will be mediated by AI assistants.
- ▶ They will constitute a **repository of all human knowledge and culture**
- ▶ They will constitute a shared infrastructure
Like the Internet today.
- ▶ **These AI platform MUST be open source**
- ▶ We need a diverse set of AI assistants for the same reasons we need a free press: linguistic, cultural, & value system diversity.
- ▶ Culture & knowledge cannot be controlled by a few companies on the West Coast of the US or in China.
- ▶ **Open source AI platforms are necessary**



什么是世界模型？



Robot Demo

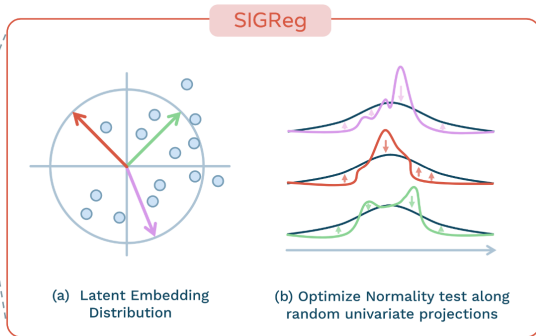
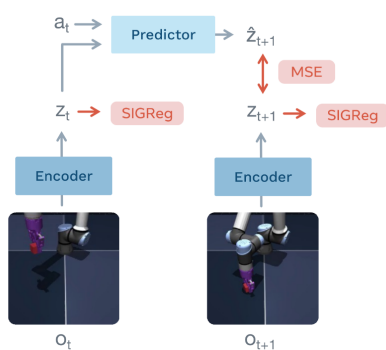


<https://ai.meta.com/vjepa/>

世界模型的应用

- ▶ 自动驾驶与无人机
- ▶ 机器人控制与具身智能
- ▶ 游戏 AI 与工业控制
- ▶ 视频生成
- ▶

训练阶段



- ▶ o_t : 时刻 t 的图像观测
- ▶ $z_t = \text{enc}_\theta(o_t)$: 编码后的隐表示
- ▶ a_t : 从 t 到 $t+1$ 的动作

$$L_{\text{pred}} = \|\hat{z}_{t+1} - z_{t+1}\|_2^2$$
$$L_{\text{LeWM}} = L_{\text{pred}} + \lambda \text{SIGReg}(Z)$$

Latent World Model

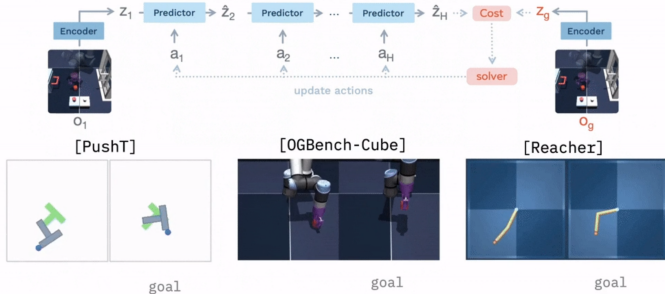
[PLDM] [-] 6 hyperparameters [-] Under-specified anti-collapse regularization	[DINO-WM] [-] Frozen Pre-trained Encoder [-] Bounded by its pretraining knowledge	[Dreamer] [-] Task specific [-] Image Reconstruction	[TD-MPC] [-] Task specific [-] State-based [-] Reward Reconstruction
[LeWM] [+] 1 hyperparameter [+] Provable anti-collapse	[LeWM] [+] End-to-End Learning	[LeWM] [+] Task agnostic [+] Reconstruction free	[LeWM] [+] Pixels Based [+] Reward-free

核心问题：如何约束 latent 学习？

- ▶ **Dreamer / TD-MPC:**
loss $\approx$ dynamics + reward + value
- ▶ **DINO-WM:**
固定表示 + loss $\approx$ dynamics, 非端到端
- ▶ **PLDM:**
端到端, loss $\approx$ dynamics, 6 个超参数
- ▶ **LeWM:**
端到端, 不引入 reward, 给出最小但有效的约束 (SIGReg)

规划阶段

Planning Phase



$$z_1 = \text{enc}(o_1), \quad z_g = \text{enc}(o_g)$$

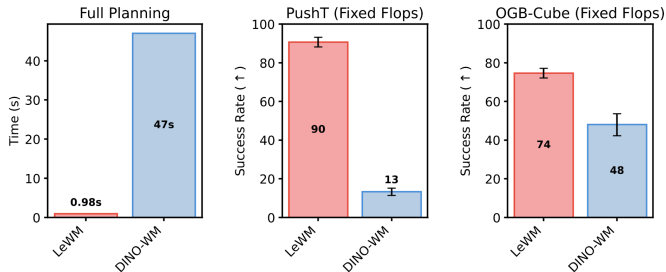
$$\hat{z}_{t+1} = \text{pred}_\phi(\hat{z}_t, a_t), \quad \hat{z}_1 = z_1$$

$$C(\hat{z}_H) = \|\hat{z}_H - z_g\|_2^2$$

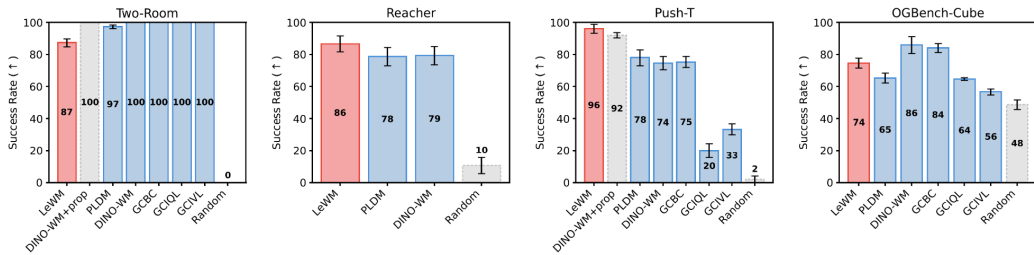
$$a_{1:H}^* = \arg \min_{a_{1:H}} C(\hat{z}_H)$$

- **Cross-Entropy Method:** 通过迭代采样与筛选 elite samples, 逐步把动作分布压向低成本区域。
- **Model Predictive Control:** 每次只执行当前规划序列的一小段, 再基于新观测继续重规划。

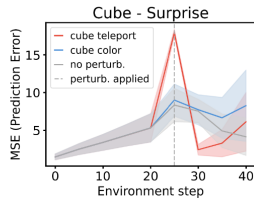
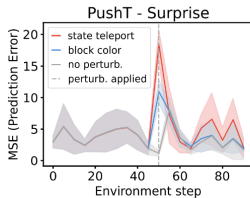
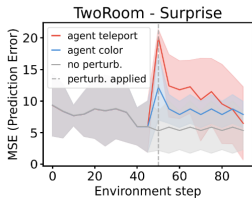
规划性能



Performance under fixed compute



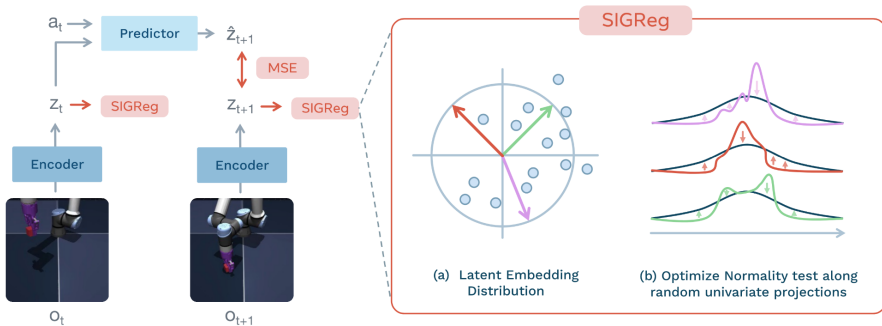
物理违背检测



$$\text{Prediction error: } ||\hat{z}_t - z||_2^2$$

- 瞬移引起的 surprise 明显高于纯变色，说明模型更敏感于时空连续性的破坏。

总结



- ▶ LeWM 将世界模型写成 latent 空间中的状态演化与规划问题。
- ▶ 它只用 prediction loss 与 SIGReg 两项损失，就实现了稳定的端到端训练。
- ▶ 相比已有 JEPA world model 路线，LeWM 更简单、更稳定，并保持了有竞争力的规划性能。